Electric Vehicle Adoption in Washington State

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Introduction

Electric vehicles (EVs) are central to cutting transport emissions, the largest source of greenhouse gases in Washington State. This project uses the **Electric Vehicle Population Data** published by the Washington State Department of Licensing on data.wa.gov (Open Database License; snapshot of 14 October 2026), which lists all 300,142 battery-electric (BEV) and plug-in hybrid (PHEV) vehicles registered with the state. Each record gives the vehicle's county, city and ZIP code, model year, make and model, EV type, electric range, eligibility for the state's clean-fuel incentive, legislative district, electric utility and census tract.

Cleaning kept the 299,303 vehicles registered in Washington with complete location fields and a model year of 2011 or later. Electric range turned out to be recorded as 0 ("not yet researched") for 66% of vehicles, almost all from model year 2021 onward, so it was not used. Features were added for region (Seattle metro, other western WA, eastern WA), primary utility, and an urbanity proxy (EVs registered in the same city). Adoption has grown steeply, peaking at 61,316 registered vehicles of model year 2023; 80.6% of vehicles are BEVs, Tesla accounts for 40.7%, and the Seattle metro (King, Snohomish and Pierce counties) holds 69% of all EVs. All analysis was done in both Python and R, with matching results.

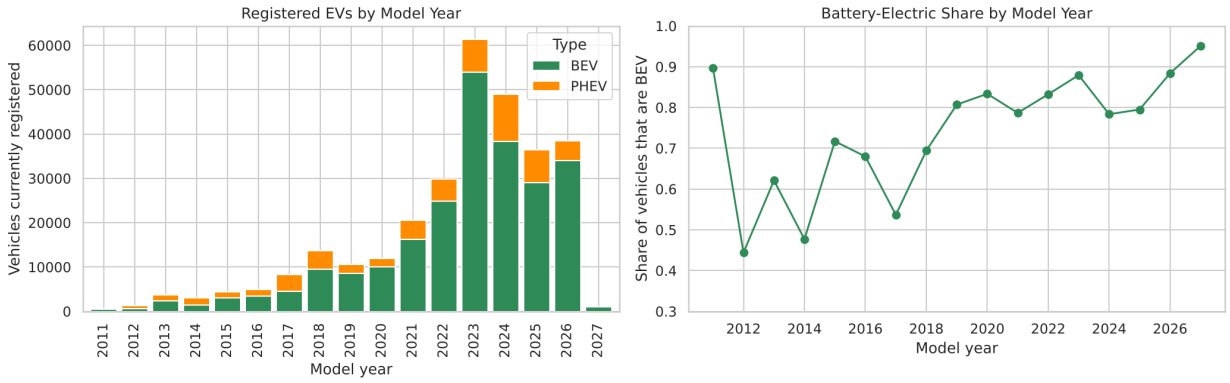


Figure 1: Registered EVs by model year (left) and the battery-electric share by model year (right).

Question 1: Can location and time predict whether an EV is battery-electric?

Introduction

A battery-electric car cuts tailpipe emissions entirely, while a plug-in hybrid still burns petrol on longer trips. If rural or older-registration areas lean towards hybrids, planners might respond with charging infrastructure or targeted incentives. This question asks whether *where and when* a vehicle is registered predicts its type (BEV or PHEV). It uses model year, region, county, city size, utility and legislative district, and deliberately excludes make, model, range and incentive eligibility, which all but define the type (make and model alone give a test AUC of 0.9997).

Approach

Three classifiers were compared: **logistic regression** (odds ratios show each factor's direction), a **decision tree** (depth 6, at least 200 vehicles per leaf) and a **random forest** (200 trees). Numeric features were standardized and categorical ones one-hot encoded. Because 80.6% of vehicles are BEVs, classes were weighted inversely to their frequency, and models were judged by **ROC-AUC**, **balanced accuracy** and the recall of each type rather than by accuracy. The data were split 80/20, stratified on the outcome, with 5-fold cross-validation on a 50,000-vehicle subsample; the random forest was trained on an 80,000-vehicle subsample to limit run time. **Permutation importance** measured each feature's contribution.

Data analysis and results

Table 1: Test-set performance (Python, threshold 0.5); the last column shows the R replication.

Model	CV ROC-AUC	Test ROC-AUC	Balanced accuracy	BEV recall	BEV precision	PHEV recall	Test ROC-AUC (R)
Logistic regression	0.616 ± 0.008	0.620	0.582	0.615	0.850	0.548	0.620
Decision tree	0.646 ± 0.008	0.649	0.604	0.641	0.860	0.568	0.639
Random forest	0.644 ± 0.010	**0.651**	**0.608**	0.656	0.861	0.560	0.651

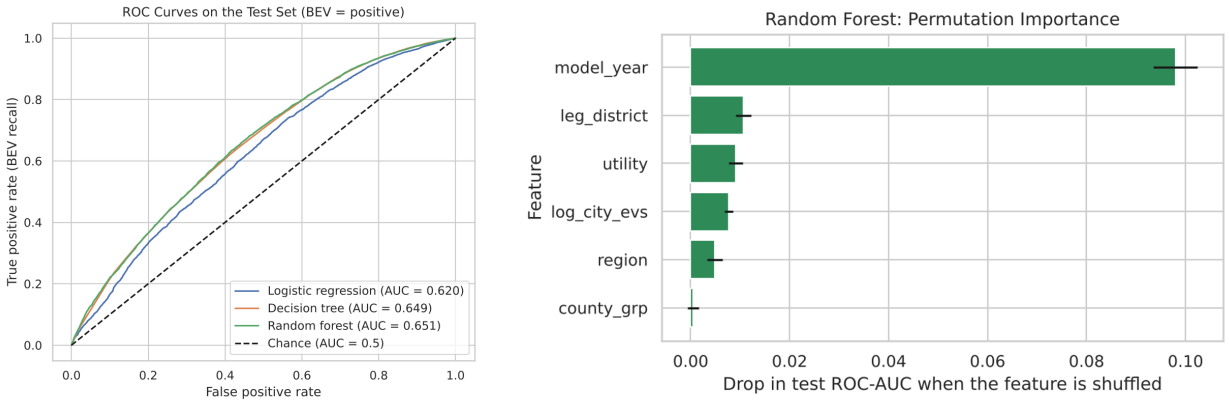


Figure 2: ROC curves on the test set (left) and the random forest's permutation importance (right).

All models beat chance but only modestly (test ROC-AUC 0.62–0.65, Table 1). The tree-based models beat logistic regression because the BEV share moves non-linearly with model year (Figure 1). **Model year** dominates: shuffling it lowers the random forest's AUC by 0.098, about ten times any location feature (Figure 2), and each standard deviation of model year (about three years) raises the odds of a BEV by 39%. Location adds little: legislative district (0.011), utility (0.009), city size (0.008) and region (0.005). The BEV share rises from 75.5% in cities with fewer than 100 EVs to 82.3% in cities with 5,000–20,000, and Seattle City Light customers have 1.36 times, and PacifiCorp customers 0.60 times, the odds of Bonneville Power Administration customers of driving a BEV.

Discussion

When a vehicle was built matters far more than where it is registered. The urban–rural gap is real but small: rural and eastern buyers lean slightly towards plug-in hybrids, consistent with longer trips and sparser charging, but location cannot reliably tell a BEV from a PHEV. The choice is driven mainly by the models available in a given year and by brand. The two languages agree on performance and on the ranking of features. One difference is that R's unpenalized logistic regression attributes the region effect to the overlapping county terms (Seattle metro odds ratio 1.05 against 1.29 in Python). Limitations: the data count vehicles currently registered rather than sales, city size is derived from the EV data itself, and some models were trained on subsamples.

Question 2: What EV-adoption profiles exist across Washington ZIP codes?

Introduction

Adoption is uneven across the state, and utilities and planners need to know *what kind* of EV market each area is, not only how many EVs it has. This question groups ZIP codes by the make-up of their EV fleet. It uses the 338 ZIP codes with at least 50 EVs (covering 99.0% of all EVs), each described by its fleet size, BEV share, Tesla share, share of recent models and brand diversity. I am interested in it because such profiles can guide where to build which kind of charging.

Approach

k-means clustering was applied to five standardized ZIP-level features: log10 number of EVs, BEV share, Tesla share, share of model year 2023 or later, and brand diversity (Shannon entropy of makes). Standardizing gives each feature equal weight. The number of clusters was chosen with the **elbow method** and the **average silhouette width** for k = 2 to 8. The silhouette was highest at k = 2 (0.304), which only separates large

from small markets, and varied little from k = 3 to 8 (0.250–0.273). **k = 4** was chosen for interpretability, accepting a slightly lower silhouette (0.250); k = 5 split off a group of only 4 ZIP codes. Clusters were named by fixed rules so that Python and R label them the same way, and profiled by their feature means and region mix.

Data analysis and results

Table 2: The four ZIP-code profiles (k-means, k = 4). R assigns the same profile to all 338 ZIP codes.

Profile	ZIP codes	EVs	Median EVs per ZIP	BEV share	Tesla share	Model year 2023+	Share in Seattle metro
Tech-suburb Tesla hubs	54	110,724	1,902	86.1%	48.5%	67.6%	91%
Diverse urban and suburban	127	143,034	1,013	78.2%	36.0%	60.7%	54%
PHEV-leaning mixed fleets	81	26,490	206	72.4%	26.3%	58.2%	16%
Small eastern Tesla-leaning	76	15,937	146	79.3%	42.2%	55.7%	21%

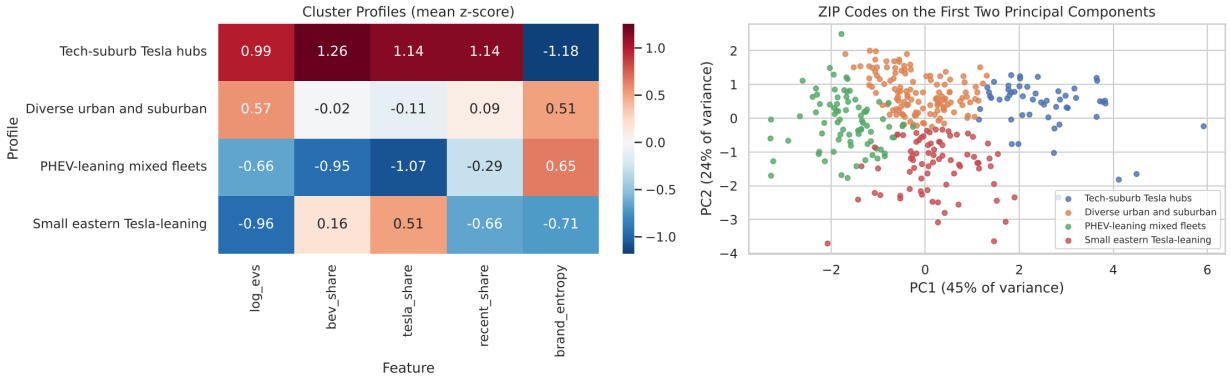


Figure 3: Mean standardized feature values per profile (left) and the ZIP codes on the first two principal components (right).

The **Tech-suburb Tesla hubs** (Redmond, Bothell, Kirkland, Bellevue, Sammamish) have the largest, newest and most Tesla-heavy fleets with the least brand variety. The **Diverse urban and suburban** group (north Seattle, Woodinville, Olympia, Camas) holds the most EVs overall, with an average BEV share and the widest mix of brands. The two small-market profiles differ sharply: **PHEV-leaning mixed fleets** (Tukwila, Bellingham, Renton, Sequim, Port Townsend) have the lowest BEV and Tesla shares, while the **Small eastern Tesla-leaning** group (Pasco, Kennewick, Moses Lake, Ellensburg; 49% in eastern Washington) combines the smallest and oldest fleets with a high Tesla share (Table 2, Figure 3).

Discussion

EV adoption is not a single urban–rural gradient. Large markets split into a premium, Tesla-dominated Eastside and a broad mainstream market, and small markets split into hybrid-leaning areas with many brands and eastern towns where a few Tesla buyers dominate, perhaps because Tesla's own fast chargers along the I-90 and I-82 corridors make a battery-electric car practical there. This suggests different priorities: more public charging where hybrids dominate, and non-Tesla fast charging in eastern towns so that other brands become viable. Python and R produced identical clusters. Limitations: the silhouette is modest, so the profiles overlap; ZIP codes vary greatly in size; and without population data the analysis describes fleet composition, not adoption rates per household.

References

Washington State Department of Licensing (2026). *Electric Vehicle Population Data*. data.wa.gov, dataset f6w7-q2d2, Open Database License (ODbL). Snapshot of 14 October 2026.

Han, J., Pei, J., & Tong, H. (2023). *Data Mining: Concepts and Techniques* (4th ed.). Morgan Kaufmann.

Code: `python/01-03_*.ipynb` and `r/01-03_*.Rmd` in the project folder reproduce every result.